

Presentation 3

Link to presentation: https://github.com/arvindkrishna87/STAT390_Winter2026/blob/master/Team2/Presentations/Slides/Presentation3_Team2.pdf

Link to code:

[Roma's code link] https://github.com/arvindkrishna87/STAT390_Winter2026/tree/master/Team2/Code/Code_Pres3_Roma

[Paige's code link - issue with branch, discussed with Noah] https://github.com/arvindkrishna87/STAT390_Winter2026/tree/main/Team2/Code/Code_Pres3_Paige_Optional

[Charlene's code link - discussed with Krish and Noah, uploaded to the main branch] https://github.com/arvindkrishna87/STAT390_Winter2026/blob/main/Team2/Code/Code_Pres3_Charlene_Visualization/Presentation3_visualization.ipynb

What did each of the team members do?

Adithi & Jason:

This week, we continued looking at why the model misclassifies many of the benign cases by systematically conducting histological analysis on half of all benign cases (14 in total) and noting common features among those misclassified. Each of us analyzed the benign cases independently, looking for patterns in tissue and cytological atypia across misclassified cases and grading each case as easy, medium, or hard difficulty to classify; we then came together and discussed our analyses for each case, putting the consensus on a spreadsheet linked below. We also worked together to synthesize our observations into overarching consensus findings about features of misclassified benign cases and recommendations for model improvement.

Link to histological analysis spreadsheet: https://docs.google.com/spreadsheets/d/1dAmrhE7B4mcspomPM3FjdsKIRN_sW6eSWxDRY4OOgHk/edit?gid=613069880#gid=613069880

Roma & Paige & Charlene:

[Roma] This week, I worked on updating the model based on the mandatory requirements. I removed dropout from the attention MLP for more stable attention weight computation, and replaced the standard attention mechanism with gated attention, which uses a tanh branch and sigmoid gate multiplied together for more expressive patch selection. I also added entropy regularization to the training loss to

prevent attention from collapsing onto just a few patches, and increased the minimum epochs before early stopping from 10 to 30 to give the model more time to converge.

[Charlene] This week, I worked on visualizing the model to understand how the mandatory changes have affected the predictions. I also determined the patches with high effective attention scores for the misclassified benign cases to understand how those were distributed. I worked with Paige to understand how the percentage predictions for benign and high-grade were split for misclassified benign cases, where I deep-dived into the patches with the highest percentage of high-grade to see where the model decided to focus more.

[Paige] I worked on implementing the code for Optional Task 3 which was obtaining patch-level predictions from the trained MIL model. I added functions to `models.py`, `trainer.py`, and `main.py` to apply the trained classifier weights directly to individual patch embeddings after training is complete. The code produces probabilities and predictions for each patch for all the test cases (with the mandatory split) which were then aggregated to the case level to report the percentage of patches predicted benign vs high grade (joined alongside the model's case prediction and true label). I also then filtered these results to identify just the false positive cases of interest - benign cases misclassified as high grade. I shared these results to Charlene and worked with her on accessing the results and making suggestions for preliminary analysis and interpreting the results.

How does it help the project?

Adithi & Jason:

Looking at the histology of misclassified benign cases helps us identify which cytologic patterns, or features of the patterns, in H&E (plus confirmatory Melan-A and SOX10 cues) reliably separate benign from high-grade cases, and which patterns are misleading the model. By pinpointing recurring misclassification drivers—like basal-layer melanosis, nuclear hyperchromasia within or above the basal layer, and goblet cell hyperplasia and potential invagination—we can refine what the model should attend to versus ignore. The added feature of goblet cell hyperplasia expands the Winter 2025 feature set and gives us another concrete explanation for a subset of misclassification errors for benign cases

Roma & Paige & Charlene:

[Paige] Obtaining the patch-level predictions hopes to investigate the model's false positive mistakes. Instead of just knowing the case was misclassified, this allows us visibility into what proportion of patches for a case may be looking high-grade to the model and furthermore which patches specifically could be driving an incorrect prediction. This can help us form hypotheses about the model errors and supports Charlene's analysis, to help improve the model and its precision.

[Charlene] Looking closer at the misclassified benign probability split between high-grade and benign. This allows us to take a look at the images and understand what the model is focusing on for misclassifying them.

[Roma] These changes improve the model's ability to identify the most relevant patches in pathology images. Gated attention gives the model more expressive power and entropy regularization prevents the model from over-relying on a single patch or slice. Together, these changes should lead to better generalization and more reliable classification performance.

Issues faced that haven't been resolved:

Adithi & Jason:

H&E is inherently coarse, so some observations are prone to interpretation differences. We also don't know which patches are considered high-attention patches by the model, so it is hard to form a holistic diagnosis as subregions may over-influence attention even when the broader tissue context suggests benignity. It is also hard to identify the presence of artifacts and nonspecific staining from the slices.

Roma & Paige & Charlene:

There's a limitation by nature in the approach and bypassing the attention in that we are looking at a patch individually with no context, which doesn't reflect what the model necessarily considers during case-level prediction - so this is just something to keep in mind when we interpret how much and what our results can say. Otherwise, we didn't have issues in our code this week, more just considering next steps.

Proposed next steps:

Adithi & Jason:

- Mark up the key features across remaining benign cases and compile examples to present alongside takeaways, focusing on goblet cell hyperplasia, pigment

distribution, nuclear hyperchromasia, melanocyte clustering, and other structural atypia by region.

- Audit current high-attention patches to test whether basal-layer melanosis is truly driving lower accuracy and identify which other “concerning” features are biasing predictions.

Roma & Paige & Charlene:

- To better understand why benign patches are being misclassified as high-grade, analyze the relationship between patch effective attention weights and the patch probabilities/predictions
 - Cross-reference patch-level high-grade predictions with attention scores to identify whether misclassified patches are also high attention, determining if the model is actively focusing on the misleading patches or if there is another influence on case prediction
 - Visual inspection of the high probability high-grade patches in the false positives
- Can also experiment with different classification thresholds beyond 0.5
- Investigate whether certain stains contribute differentially to false positive patch predictions
- Run same analysis on validation set for more examples and to see if there are patterns

References

- <https://www.sciencedirect.com/science/article/pii/S1361841521000785>
- [Fall 2025 Presentation 4](#)
- [Winter 2026 Presentation 6](#)
- Butt, K.; Hussain, R.; Coupland, S.E.; Krishna, Y. Conjunctival Melanoma: A Clinical Review and Update. *Cancers* 2024, 16, 3121. <https://doi.org/10.3390/cancers16183121>
- [https://www.pathologyoutlines.com/topic/eyepamconj.html#:~:text=Primary%20acquired%20melanosis%20\(PAM\)%20is%20also%20known,is%20significant%20factor%20for%20recurrence%20and%20progression](https://www.pathologyoutlines.com/topic/eyepamconj.html#:~:text=Primary%20acquired%20melanosis%20(PAM)%20is%20also%20known,is%20significant%20factor%20for%20recurrence%20and%20progression)
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- <https://pmc.ncbi.nlm.nih.gov/articles/PMC11429624/>
- [Winter 2025 Presentation 3](#)
- [Winter 2025 Presentation 4](#)
- [Winter 2025 Presentation 5](#)